

Администрирование локальных сетей

Отчёт по лабораторной работе 8. Настройка сетевых сервисов. DHCP

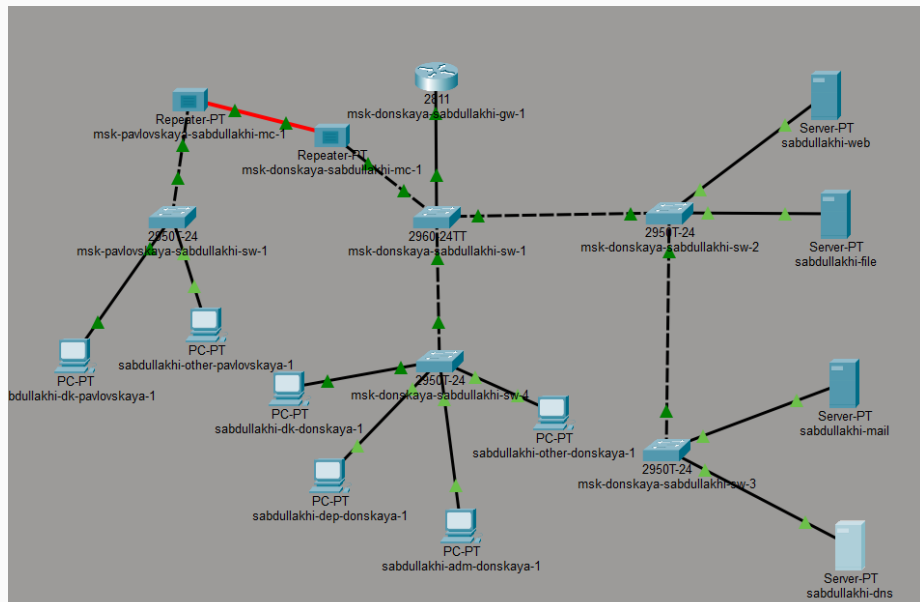
Абдуллахи Шугофа

6 April 2026

Российский университет дружбы народов, Москва, Россия

Освоить на практике механизм динамического назначения IP-адресов через протокол DHCP (Dynamic Host Configuration Protocol) в локальной вычислительной сети, а также получить навыки настройки DNS-сервера и его взаимодействия с DHCP.

Построение топологии сети



Настройка IP-адреса на DNS-сервере вручную

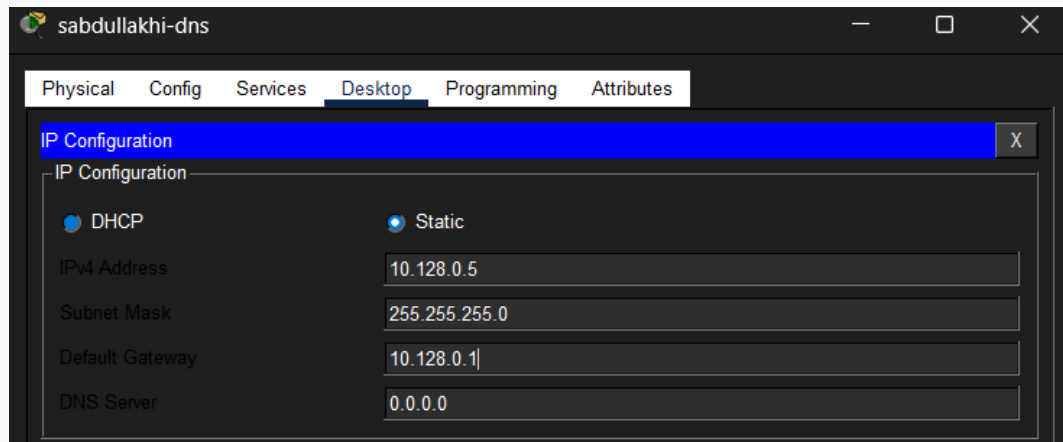


Рис. 2: Настройка IP-адреса DNS-сервера

The screenshot shows the Mikrotik WinBox interface with the 'Services' tab selected. The 'DNS' service is enabled, indicated by a blue dot. Below the service list, there are buttons for 'Add', 'Save', and 'Remove'. The 'A Records' table is visible, showing four entries:

No.	Name	Type	Detail
0	dns.donskaya.rudn.ru	A Record	10.128.0.5
1	file.donskaya.rudn.ru	A Record	10.128.0.3
2	mail.donskaya.rudn.ru	A Record	10.128.0.4
3	www.donskaya.rudn.ru	A Record	10.128.0.2

Настройка DHCP на маршрутизаторе

```
msk-donskaya-sabdullakhi-gw-1
Physical Config CLI Attributes
Password:
msk-donskaya-sabdullakhi-gw-1>enab
Password:
msk-donskaya-sabdullakhi-gw-1#conf term
Enter configuration commands, one per line. End with CNIL/Z.
msk-donskaya-sabdullakhi-gw-1(config)#ip name-server 10.128.0.5
msk-donskaya-sabdullakhi-gw-1(config)#server dhcp
^
% Invalid input detected at '^' marker.

msk-donskaya-sabdullakhi-gw-1(config)#service dhcp
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp pool dk
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#network 10.128.3.0 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#default-router 10.128.3.1
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#exit
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.3.200 10.128.3.254
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp pool departments
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#network 10.128.4.0 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#default-router 10.128.4.1
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#exit
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp pool adm
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#network 10.128.5.0 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#default-router 10.128.5.1
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#exit
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp pool other
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#network 10.128.6.0 255.255.255.0
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#default-router 10.128.6.1
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-sabdullakhi-gw-1(dhcp-config)#exit
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.6.1 10.128.6.29
msk-donskaya-sabdullakhi-gw-1(config)#ip dhcp excluded-address 10.128.6.200 10.128.6.254
msk-donskaya-sabdullakhi-gw-1(config)#exit
msk-donskaya-sabdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Получение IP-адресов клиентами по DHCP

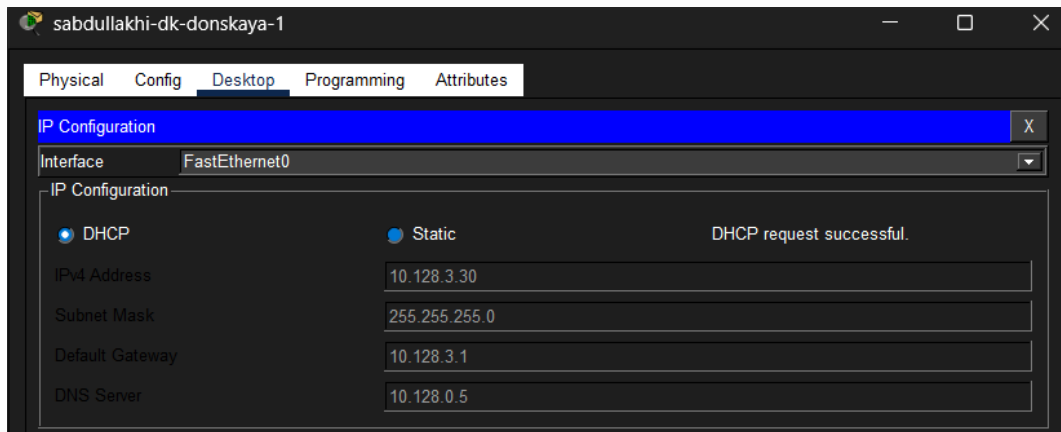
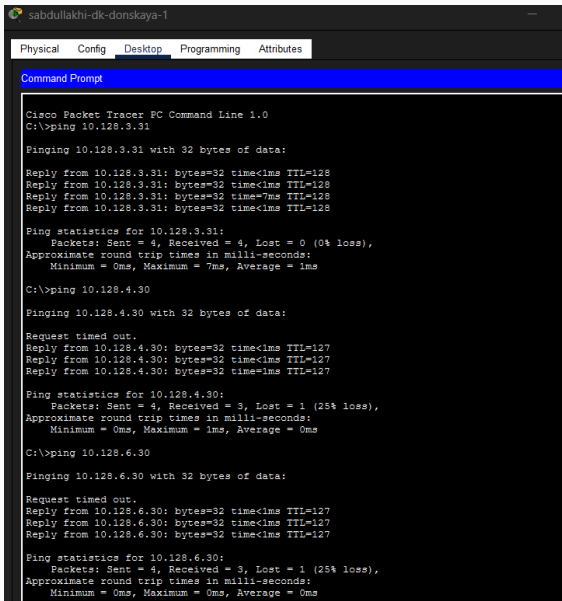


Рис. 5: Получение IP-адресов клиентами по DHCP

Проверка сетевой доступности (связности)



```
sabdullakhi-dk-donskaya-1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.31

Pinging 10.128.3.31 with 32 bytes of data:

Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time=7ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.31:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>ping 10.128.4.30

Pinging 10.128.4.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.4.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.6.30

Pinging 10.128.6.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.6.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```


Просмотр информации о пулах DHCP на маршрутизаторе

```
msk-donskaya-sabdullakhi-gw-1#sh ip dhcp pool

Pool dk :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 2
  Excluded addresses                : 8
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  10.128.3.1         10.128.3.1 - 10.128.3.254  2 / 8 / 254

Pool departments :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 1
  Excluded addresses                : 8
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  10.128.4.1         10.128.4.1 - 10.128.4.254  1 / 8 / 254

Pool adm :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 1
  Excluded addresses                : 8
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  10.128.5.1         10.128.5.1 - 10.128.5.254  1 / 8 / 254

Pool other :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 2
  Excluded addresses                : 8
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  10.128.6.1         10.128.6.1 - 10.128.6.254  2 / 8 / 254
```

Просмотр информации о пулах DHCP на маршрутизаторе

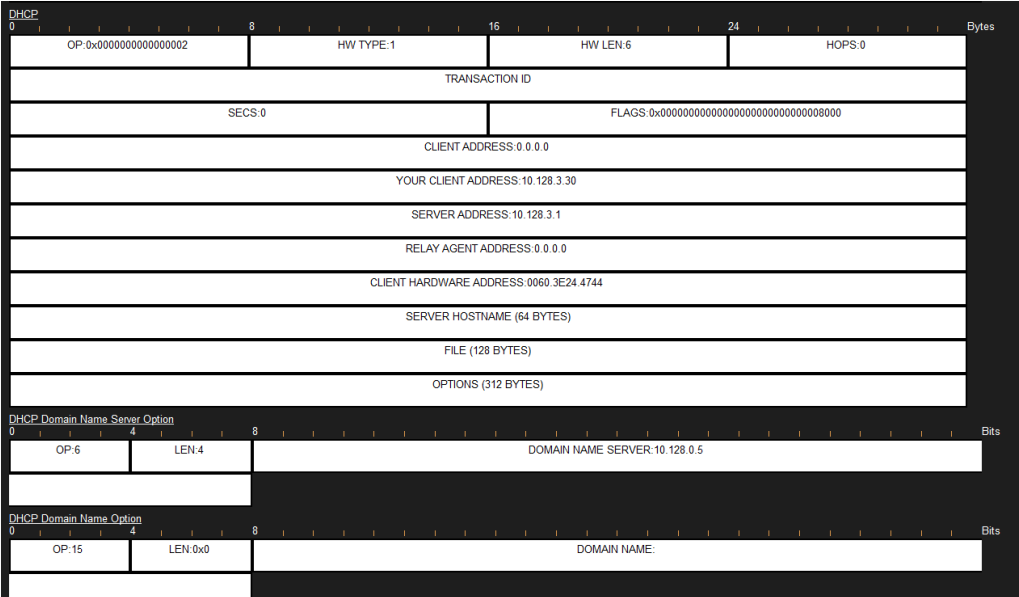
```
msk-donskaya-sabdullakhi-gw-1#sh ip dhcp binding
IP address      Client-ID/      Lease expiration    Type
                Hardware address
10.128.3.30      0060.3E24.4744  --                  Automatic
10.128.3.31      00E0.F750.9E45  --                  Automatic
10.128.4.30      0030.A3D6.C429  --                  Automatic
10.128.5.30      0050.0FE1.7062  --                  Automatic
10.128.6.30      0004.9A7B.475D  --                  Automatic
10.128.6.31      0060.5CD0.EE8B  --                  Automatic
msk-donskaya-sabdullakhi-gw-1#
```

Рис. 8: Таблица выданных DHCP-адресов

Анализ процесса обмена DHCP-сообщениями в режиме симуляции

[illegible]

DHCP Offer



DHCP Request

PDU Information at Device: msk-donskaya-sabdullakhi-sw-1

OSI Model

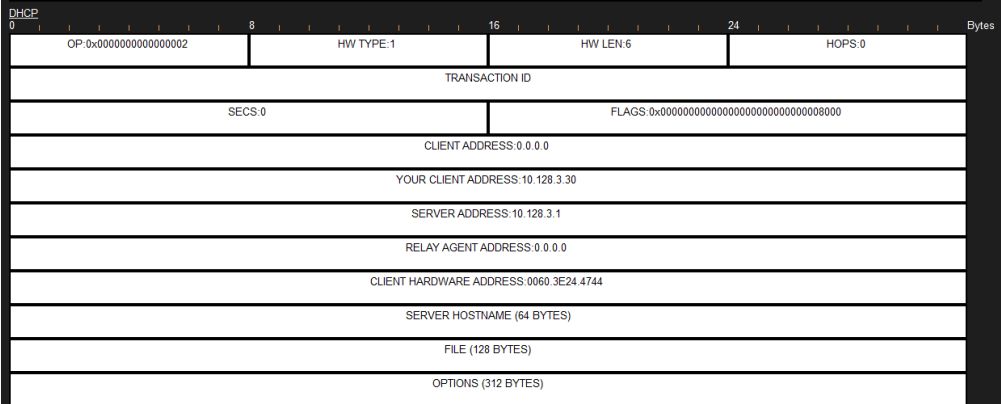
Inbound PDU Details

Outbound PDU Details

DHCP

0	8	16	24	Bytes
OP:0x0000000000000002	HW TYPE:1	HW LEN:6	HOPS:0	
TRANSACTION ID				
SECS:0		FLAGS:0x00000000000000000000000000008000		
CLIENT ADDRESS:0.0.0.0				
YOUR CLIENT ADDRESS:10.128.3.30				
SERVER ADDRESS:10.128.3.1				
RELAY AGENT ADDRESS:0.0.0.0				
CLIENT HARDWARE ADDRESS:0060.3E24.4744				
SERVER HOSTNAME (64 BYTES)				
FILE (128 BYTES)				

DHCP Acknowledgment (ACK)



Процесс обмена DHCP-сообщениями в режиме симуляции

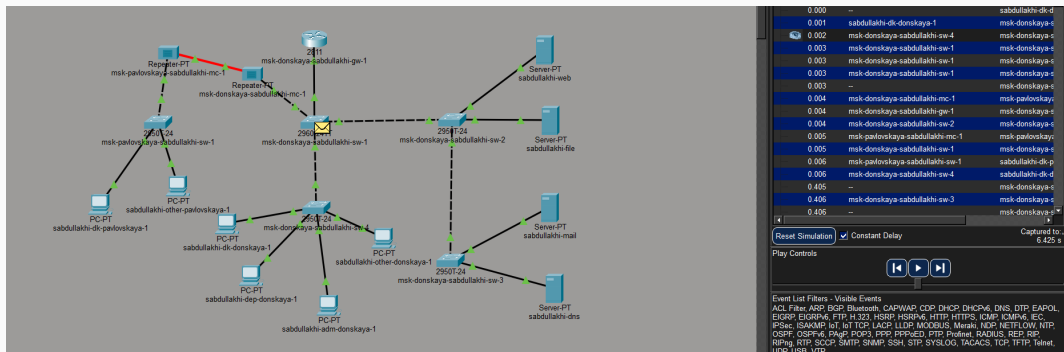


Рис. 13: Процесс обмена DHCP-сообщениями в режиме симуляции

В ходе лабораторной работы был настроен DNS-сервер и созданы A-записи для четырёх доменных имён. На маршрутизаторе настроен DHCP-сервис с четырьмя пулами адресов для разных подсетей, клиенты успешно получили сетевые параметры автоматически. Проверка связности подтвердила корректную работу маршрутизации, а в режиме симуляции изучен полный процесс обмена DHCP-сообщениями (Discover, Offer, Request, ACK). Цель работы достигнута.

Спасибо за внимание
